Linear Algebra and Its Applications

Section 4.6: Rank

Detailed worked solutions

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Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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4.6 Rank

For an $m \times n$ matrix, the rank r is the number of pivots, the dimension of its column space, and the dimension of its row space. The rank–nullity equation is

$$\dim \operatorname{Nul} A = n - r.$$

Notice the number of *columns* on the right. Row operations preserve the row space and null space, but may change the column space. Consequently use pivot columns of the *original* matrix for a column-space basis; use nonzero rows of an echelon form for a row-space basis. A basis for a null space comes from solving the homogeneous equations with free variables.

Practice problem 4.6.P1: Rank and nullity from the given echelon form

The given echelon matrix B has two nonzero rows and two pivot positions. Thus $\operatorname{rank} A = 2$. Since A has five columns, rank–nullity gives $\dim \operatorname{Nul} A = 5 - 2 = 3$.

Practice problem 4.6.P2: Choose the correct rows and columns

The pivot columns of B are columns $1, 2$. Therefore a column-space basis uses the original columns $(2, 1, -7, 4)$ and $(-1, -2, 8, -5)$ of A. A row-space basis uses the two nonzero rows of B:

$$(1, -2, -4, 3, -2), \qquad (0, 3, 9, -12, 12).$$

The former vectors have four entries and live in $\mathbb{R}^4$; the latter have five entries and live in $\mathbb{R}^5$. This is a useful check on their roles.

Practice problem 4.6.P3: The next step toward a null-space basis

Continue reducing B to reduced echelon form: divide its second row by 3, then add twice that new row to its first. The nonzero rows become $(1, 0, 2, -5, 6)$ and $(0, 1, 3, -4, 4)$. From there take x_3, x_4, x_5 free, express x_1, x_2 in terms of them, and separate their coefficient vectors. It is the free variables, not the pivot columns, that directly produce this basis.

Practice problem 4.6.P4: Transpose does not change rank

The rank theorem gives $\operatorname{rank} A^T = \operatorname{rank} A = 2$. Therefore every row echelon form of A^T has exactly two pivot columns.

Exercise 4.6.1: Three bases from the supplied echelon form

The supplied B has 2 pivots, so $\operatorname{rank} A = 2$ and $\dim \operatorname{Nul} A = 4 - 2 = 2$, without row reducing A. Its pivot columns are 1, 2.

Column and row bases use different source matrices.

A matrix whose columns form a basis of $\operatorname{Col} A$ is

$$C = \begin{pmatrix} 1 & -4 \\ -1 & 2 \\ 5 & -6 \end{pmatrix}.$$

These are the pivot columns of A, not B. A matrix whose rows form a basis of $\operatorname{Row} A$ is

$$R = \begin{pmatrix} 1 & 0 & -1 & 5 \\ 0 & -2 & 5 & -6 \end{pmatrix}.$$

Solve the homogeneous equations for the free variables.

Back substitution in $Bx = 0$ gives a basis matrix

$$N = \begin{pmatrix} 2 & -5 \\ 5 & -3 \\ 2 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \operatorname{Nul} A = \{Nz : z \in \mathbb{R}^2\}.$$

The columns of N are the coefficient vectors of the free parameters (some have been multiplied by nonzero scalars to avoid fractions). Each solves $Bx = 0$, and hence $Ax = 0$. They are independent because their free-coordinate entries isolate the parameters; their number equals the number of free variables. Thus they span the entire null space, not only some solutions.

Exercise 4.6.2: Three bases from the supplied echelon form

The supplied B has 3 pivots, so $\operatorname{rank} A = 3$ and $\dim \operatorname{Nul} A = 5 - 3 = 2$, without row reducing A. Its pivot columns are 1, 3, 5.

Column and row bases use different source matrices.

A matrix whose columns form a basis of $\operatorname{Col} A$ is

$$C = \begin{pmatrix} 1 & 4 & 2 \\ 2 & 6 & -3 \\ 3 & 3 & -3 \\ 3 & 0 & 0 \end{pmatrix}.$$

These are the pivot columns of A, not B. A matrix whose rows form a basis of $\operatorname{Row} A$ is

$$R = \begin{pmatrix} 1 & 3 & 4 & -1 & 2 \\ 0 & 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 & -5 \end{pmatrix}.$$

Solve the homogeneous equations for the free variables.

Back substitution in $Bx = 0$ gives a basis matrix

$$N = \begin{pmatrix} -3 & -3 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad \operatorname{Nul} A = \{Nz : z \in \mathbb{R}^2\}.$$

The columns of N are the coefficient vectors of the free parameters (some have been multiplied by nonzero scalars to avoid fractions). Each solves $Bx = 0$, and hence $Ax = 0$. They are independent because their free-coordinate entries isolate the parameters; their number equals the number of free variables. Thus they span the entire null space, not only some solutions.

Exercise 4.6.3: Three bases from the supplied echelon form

The supplied B has 3 pivots, so $\operatorname{rank} A = 3$ and $\dim \operatorname{Nul} A = 6 - 3 = 3$, without row reducing A. Its pivot columns are 1, 2, 5.

Column and row bases use different source matrices.

A matrix whose columns form a basis of $\operatorname{Col} A$ is

$$C = \begin{pmatrix} 2 & 6 & 3 \\ -2 & -3 & 0 \\ 4 & 9 & 3 \\ -2 & 3 & 3 \end{pmatrix}.$$

These are the pivot columns of A, not B. A matrix whose rows form a basis of $\operatorname{Row} A$ is

$$R = \begin{pmatrix} 2 & 6 & -6 & 6 & 3 & 6 \\ 0 & 3 & 0 & 3 & 3 & 0 \\ 0 & 0 & 0 & 0 & 3 & 0 \end{pmatrix}.$$

Solve the homogeneous equations for the free variables.

Back substitution in $Bx = 0$ gives a basis matrix

$$N = \begin{pmatrix} 3 & 0 & -3 \\ 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \operatorname{Nul} A = \{Nz : z \in \mathbb{R}^3\}.$$

The columns of N are the coefficient vectors of the free parameters (some have been multiplied by nonzero scalars to avoid fractions). Each solves $Bx = 0$, and hence $Ax = 0$. They are independent because their free-coordinate entries isolate the parameters; their number equals the number of free variables. Thus they span the entire null space, not only some solutions.

Exercise 4.6.4: Three bases from the supplied echelon form

The supplied B has 5 pivots, so $\operatorname{rank} A = 5$ and $\dim \operatorname{Nul} A = 6 - 5 = 1$, without row reducing A. Its pivot columns are 1, 2, 3, 5, 6.

Column and row bases use different source matrices.

A matrix whose columns form a basis of $\operatorname{Col} A$ is

$$C = \begin{pmatrix} 1 & 1 & -2 & 1 & -2 \\ 1 & 2 & -3 & -2 & -3 \\ 1 & -1 & 0 & 1 & 6 \\ 1 & -2 & 2 & -3 & 0 \\ 1 & -2 & 1 & 2 & -1 \end{pmatrix}.$$

These are the pivot columns of A, not B. A matrix whose rows form a basis of $\operatorname{Row} A$ is

$$R = \begin{pmatrix} 1 & 1 & -2 & 0 & 1 & -2 \\ 0 & 1 & -1 & 0 & -3 & -1 \\ 0 & 0 & 1 & 1 & -13 & -1 \\ 0 & 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

Solve the homogeneous equations for the free variables.

Back substitution in $Bx = 0$ gives a basis matrix

$$N = \begin{pmatrix} -1 \\ -1 \\ -1 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \qquad \operatorname{Nul} A = \{Nz : z \in \mathbb{R}^1\}.$$

The columns of N are the coefficient vectors of the free parameters (some have been multiplied by nonzero scalars to avoid fractions). Each solves $Bx = 0$, and hence $Ax = 0$. They are independent because their free-coordinate entries isolate the parameters; their number equals the number of free variables. Thus they span the entire null space, not only some solutions.

Exercise 4.6.5: A rank-three matrix with seven columns

Rank–nullity gives $\dim \operatorname{Nul} A = 7 - 3 = 4$. The row-space dimension equals rank, so $\dim \operatorname{Row} A = 3$. Transpose preserves rank, so $\operatorname{rank} A^T = 3$.

Exercise 4.6.6: A rank-two matrix with five columns

There are $5 - 2 = 3$ nonpivot columns, giving $\dim \operatorname{Nul} A = 3$. The rank theorem gives $\dim \operatorname{Row} A = 2$ and $\operatorname{rank} A^T = 2$.

Exercise 4.6.7: Full row rank does not identify a null space with its coordinate space

Four pivot columns in a 4×7 matrix mean rank 4, so its columns span all of $\mathbb{R}^4$: $\operatorname{Col} A = \mathbb{R}^4$. Rank–nullity gives nullity $7 - 4 = 3$. But $\operatorname{Nul} A$ is a three-dimensional subspace of $\mathbb{R}^7$, not the set $\mathbb{R}^3$ of triples. Equality $\operatorname{Nul} A = \mathbb{R}^3$ is therefore false; a choice of basis gives an isomorphism with $\mathbb{R}^3$, not literal equality of the sets.

Exercise 4.6.8: Rank four with six output coordinates

The nullity is $8-4=4$. The column space is a four-dimensional subspace of $\mathbb{R}^6$, because the columns have six entries. Thus it is not literally $\mathbb{R}^4$; dimension does not change the number of entries in its vectors.

Exercise 4.6.9: Recover rank from a known nullity

The matrix has six columns and nullity three, so its rank is $6-3=3$. Thus $\dim \operatorname{Col} A = 3$. But the columns have four entries, so $\operatorname{Col} A$ is a subspace of $\mathbb{R}^4$, not literally $\mathbb{R}^3$.

Exercise 4.6.10: Seven columns and five free coordinates

By rank–nullity, $\operatorname{rank} A = 7-5=2$. Column-space dimension equals rank, so the requested dimension is 2. The eight rows specify the ambient output space but do not replace the seven in the rank–nullity equation.

Exercise 4.6.11: From nullity to row-space dimension

Here $n=5$ and nullity is 3, so rank is $5-3=2$. The row-space dimension equals rank and is therefore 2.

Exercise 4.6.12: A second row-space dimension

There are four columns and nullity 2. Hence rank, and therefore row-space dimension, equals $4-2=2$.

Exercise 4.6.13: The largest possible rank

There can be no more pivots than rows or columns, so an $m \times n$ matrix has rank at most $\min(m,n)$. Both a 7×5 and a 5×7 matrix therefore have largest possible rank 5. This is attainable by placing a 5×5 identity matrix in the first five rows or columns and filling remaining entries with zeros.

Exercise 4.6.14: The largest row-space dimension

The row-space dimension is rank. For sizes 5×4 and 4×5, the maximum rank is 4 in each case. Identity blocks with an additional zero row or zero column show that this bound is attained.

Exercise 4.6.15: The smallest nullity for a wide matrix

A 3×7 matrix has rank at most 3, so its nullity is $7-\operatorname{rank} A \geq 7-3=4$. The matrix $[\,I_3\ 0\,]$ attains rank 3 and nullity 4. Thus the smallest possible nullity is 4.

Exercise 4.6.16: The smallest nullity for a tall matrix

A 7×5 matrix can have rank 5, so its nullity can be $5 - 5 = 0$. For example stack two zero rows below I_5. Nullity cannot be negative, so zero is the minimum.

Exercise 4.6.17: Five rank statements

(a) **True.** The columns of A^T are exactly the transposed rows of A; their spans are the same coordinate vectors.

(b) **False.** The first original rows need not be independent. For $A = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ there are three pivots but the first three rows are dependent.

(c) **True.** Both dimensions equal rank, including for rectangular matrices.

(d) **False.** The sum equals the number of *columns*, n, not the number of rows m.

(e) **True.** Finite-precision rounding can turn an exact zero pivot into a small nonzero value or the reverse. Exact elementary operations preserve rank; its apparent computed value may change, as the Numerical Note explains.

Exercise 4.6.18: Five statements about row operations

(a) **False.** Use pivot columns of A for $\operatorname{Col} A$, not generally those of its echelon form. For example swapping the rows of $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ changes its column span.

(b) **False.** The row span is preserved, but the particular relations among the rows can change. Rows $(1, 0), (1, 0)$ satisfy $r_2 - r_1 = 0$; a row replacement gives rows $(1, 0), (0, 0)$, for which that relation no longer holds.

(c) **True.** Nullity equals the number of free variables, namely the number of nonpivot columns.

(d) **True.** Transposing the rows of A^T gives the columns of A.

(e) **True.** Every new row is a combination of old rows, and invertibility of elementary operations gives the reverse inclusion of row spans.

Exercise 4.6.19: A one-dimensional kernel of a five-row system

All homogeneous solutions are multiples of one nonzero solution, so nullity is 1. With six columns, rank is $6 - 1 = 5$, equal to the number of rows. Thus there is a pivot in every row, and $Ax = b$ is consistent for every right-hand side $b \in \mathbb{R}^5$. The answer is yes.

Exercise 4.6.20: Two free variables in an eight-variable system

A consistent system with eight variables and two free variables has six pivots. There are also six equations, so every row has a pivot. Therefore $Ax = b$ is consistent for every $b \in \mathbb{R}^6$; changing constants alone cannot make it inconsistent. The answer is no.

Exercise 4.6.21: Onto with just one free direction

Consistency for every right-hand side in a nine-row system means rank 9. With ten variables the nullity is $10 - 9 = 1$. Thus all homogeneous solutions are multiples of one vector. It is impossible to find two nonzero homogeneous solutions that are not multiples of each other.

Exercise 4.6.22: Too many variables for a one-dimensional kernel

For ten equations in twelve variables, rank is at most ten. Nullity is therefore at least $12 - 10 = 2$. It cannot be 1, so not all homogeneous solutions can be multiples of one fixed nonzero vector.

Exercise 4.6.23: Replace redundant equations

The two given nonproportional solutions are independent and span the kernel, so nullity is 2. There are eight variables, giving rank $8 - 2 = 6$. Row reduction therefore leaves six nonzero rows and six redundant zero rows. Keeping just the six nonzero homogeneous equations describes exactly the same solutions. Fewer than six cannot suffice: a matrix with fewer rows would have rank below six and nullity above two.

Exercise 4.6.24: A unique solution for one right-hand side versus all

A 7×6 matrix may have rank 6, giving zero kernel. Then every consistent system has a unique solution; for example $A = \begin{pmatrix} I_6 \\ 0 \end{pmatrix}$ and $b = 0$ do. But rank is at most $6 < 7$, so the column space cannot be all of $\mathbb{R}^7$. Thus a unique solution for some right-hand side is possible, while a unique solution for every right-hand side is impossible.

Exercise 4.6.25: One missing pivot row leaves some constants unattainable

Twelve variables and three free variables mean rank 9. The system has ten rows, so it lacks a pivot in one row. Its column space is a proper subspace of $\mathbb{R}^{10}$, and some right-hand sides lie outside it. Thus the scientist cannot guarantee a solution after changing the constants; indeed there necessarily exist choices of constants that make the system inconsistent.

Exercise 4.6.26: Full rank for a tall matrix

When $m > n$, the largest possible rank is n. The columns are independent if and only if all n columns are pivot columns, which is equivalent to rank n. This is precisely full rank for the given shape of matrix.

Exercise 4.6.27: Locate the four fundamental spaces

Rows of A and inputs to A have n entries; columns of A and inputs to A^T have m entries. Consequently

$$\begin{array}{c|cc} \mathbb{R}^n & \operatorname{Row} A = \operatorname{Col} A^T, & \operatorname{Nul} A \\ \mathbb{R}^m & \operatorname{Col} A = \operatorname{Row} A^T, & \operatorname{Nul} A^T \end{array}$$

The six names reduce to four fundamental spaces after the two displayed identifications. For special matrices some of these four may coincide as sets (for example several can be the zero space); "four" refers to the four roles, not a guarantee of four different sets in every example.

Exercise 4.6.28: Two rank–nullity identities

(a) Apply rank–nullity to A.

Since $\dim \operatorname{Row} A = \operatorname{rank} A$ and $\dim \operatorname{Nul} A = n - \operatorname{rank} A$, their sum is n.

(b) Apply rank–nullity to A^T.

This transpose has m columns and rank equal to that of A. Thus $\dim \operatorname{Nul} A^T = m - \operatorname{rank} A$; adding $\dim \operatorname{Col} A = \operatorname{rank} A$ gives m.

Exercise 4.6.29: Onto is equivalent to a trivial transpose kernel

The equation $Ax = b$ is solvable for every $b \in \mathbb{R}^m$ if and only if $\dim \operatorname{Col} A = m$. By Exercise 4.6.28(b), this is equivalent to $\dim \operatorname{Nul} A^T = 0$, which is equivalent to $\operatorname{Nul} A^T = \{0\}$. That last condition says exactly that $A^T x = 0$ has only the trivial solution.

Exercise 4.6.30: The augmented-rank consistency criterion

Adding b as a new column increases rank precisely when b is outside the span of the old columns. Therefore $Ax = b$ is consistent if and only if

$$\operatorname{rank}[\, A \;\; b \,] = \operatorname{rank} A.$$

Equivalently, an inconsistent row in the reduced augmented matrix gives an extra pivot in its last column. If ranks agree, that extra pivot is absent.

Exercise 4.6.31: An outer product has at most one independent column

For $u = (2, -3, 5)$ and $v = (a, b, c)$,

$$uv^T = \begin{pmatrix} 2a & 2b & 2c \\ -3a & -3b & -3c \\ 5a & 5b & 5c \end{pmatrix}.$$

Its columns are au, bu, cu, so its column space is contained in $\operatorname{Span}\{u\}$, a one-dimensional space. Hence rank is at most 1. It is zero when $v = 0$, and otherwise is exactly 1, since $u \neq 0$.

Exercise 4.6.32: Factor a rank-one matrix

With $u = (1, 2)$ the matrix uv^T has first row v^T and second row $2v^T$. Thus choose $v = (1, -3, 4)$, giving

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} [\, 1 \;\; -3 \;\; 4 \,] = \begin{pmatrix} 1 & -3 & 4 \\ 2 & -6 & 8 \end{pmatrix}.$$

Exercise 4.6.33: Every nonzero column generates a rank-one column space

If rank is 1 and the first column u is nonzero, it is a basis for the column space. Thus the three columns have form $u, \alpha u, \beta u$. Taking $v = (1, \alpha, \beta)$ gives $A = uv^T$. If the first column is zero, choose any other nonzero column as u (one exists because rank is 1). Express each column as $v_j u$, including coefficient zero for a zero column, and set $v = (v_1, v_2, v_3)$. The same outer-product formula then holds.

Exercise 4.6.34: A rank-r matrix is a sum of r rank-one matrices

Since U is obtained by elementary row operations, $U = FA$ for an invertible product F of elementary matrices. Set $E = F^{-1}$ to get $A = EU$. Write $E = [\, e_1 \ \cdots \ e_m \,]$ and let u_i^T be row i of U. Matrix multiplication by columns and rows gives

$$A = EU = \sum_{i=1}^{m} e_i u_i^T = \sum_{i=1}^{r} e_i u_i^T.$$

The last equality uses the zero rows below the first r rows of the echelon matrix. Each of those first r rows is nonzero, and each column e_i of an invertible matrix is nonzero. Thus every displayed outer product has rank exactly 1, proving the assertion.

Exercise 4.6.35: Four fundamental bases and two invertible assembled matrices

$$A = \begin{pmatrix} 7 & -9 & -4 & 5 & 3 & -3 & -7 \\ -4 & 6 & 7 & -2 & -6 & -5 & 5 \\ 5 & -7 & -6 & 5 & -6 & 2 & 8 \\ -3 & 5 & 8 & -1 & -7 & -4 & 8 \\ 6 & -8 & -5 & 4 & 4 & 9 & 3 \end{pmatrix}.$$

(a) Supply and verify the three requested bases.

The pivot columns are 1, 2, 4, 6, so rank is 4. The following matrices give respectively a column-space basis in their columns, a null-space basis in their columns, and a row-space basis in their rows:

$$C = \begin{pmatrix} 7 & -9 & 5 & -3 \\ -4 & 6 & -2 & -5 \\ 5 & -7 & 5 & 2 \\ -3 & 5 & -1 & -4 \\ 6 & -8 & 4 & 9 \end{pmatrix}.$$

$$N = \begin{pmatrix} -13 & -10 & 3 \\ -11 & -1 & -2 \\ 2 & 0 & 0 \\ 0 & 11 & -7 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \\ 0 & 0 & 1 \end{pmatrix}.$$

$$R = \begin{pmatrix} 1 & 0 & \frac{13}{2} & 0 & 5 & 0 & -3 \\ 0 & 1 & \frac{11}{2} & 0 & \frac{1}{2} & 0 & 2 \\ 0 & 0 & 0 & 1 & -\frac{11}{2} & 0 & 7 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 \end{pmatrix}.$$

The rows of R are the nonzero rows of the reduced echelon form. The 3 columns of N result from its free-variable equations; exact multiplication gives $AN = 0$, and their free coordinates show independence. Thus these are bases, not only spanning lists.

(b) Supply the transpose-kernel basis and check invertibility.

$$M = \begin{pmatrix} 2 \\ 41 \\ 0 \\ -28 \\ 11 \end{pmatrix}.$$

Here $A^T M = 0$, and its 1 columns are independent. Form $S = [\, R^T \; N \,]$ and $T = [\, C \; M \,]$ using the displayed blocks. There are $4 + 3 = 7$ columns in $S \in M_{7\times 7}$ and $4 + 1 = 5$ columns in $T \in M_{5\times 5}$. Exact determinants are

$$\det S = -\frac{88369}{2}, \qquad \det T = 10360.$$

Both are nonzero, verifying that both square matrices are invertible. Conceptually, a vector in both $\operatorname{Row} A$ and $\operatorname{Nul} A$ has form $x = A^T y$ and satisfies $Ax = 0$, so $x^T x = y^T Ax = 0$. Since $x^T x$ is a sum of squares, $x = 0$. The same argument with A^T treats $\operatorname{Col} A$ and $\operatorname{Nul} A^T$; their dimensions add to the full ambient dimensions. This explains why assembling their bases produces invertible matrices.

Exercise 4.6.36: Four fundamental bases and two invertible assembled matrices

The problem permits a generated example. Here is one reproducible sample: with Python `random.Random(46)`, draw the entries of a 6×2 matrix J and then a 2×7 matrix K successively from `randint(-1,1)`, and set $A = JK$. This guarantees rank at most 2, which in particular is at most 4. The resulting matrix is displayed below; its first two columns are independent, so its actual rank is 2.

$$A = \begin{pmatrix} 1 & 1 & 0 & -1 & 1 & 1 & 1 \\ 2 & 1 & 1 & -1 & 2 & 0 & 1 \\ -2 & -1 & -1 & 1 & -2 & 0 & -1 \\ 0 & -1 & 1 & 1 & 0 & -2 & -1 \\ 2 & 1 & 1 & -1 & 2 & 0 & 1 \\ -2 & -1 & -1 & 1 & -2 & 0 & -1 \end{pmatrix}.$$

(a) Supply and verify the three requested bases.

The pivot columns are 1, 2, so rank is 2. The following matrices give respectively a column-space

basis in their columns, a null-space basis in their columns, and a row-space basis in their rows:

$$C = \begin{pmatrix} 1 & 1 \\ 2 & 1 \\ -2 & -1 \\ 0 & -1 \\ 2 & 1 \\ -2 & -1 \end{pmatrix}.$$

$$N = \begin{pmatrix} -1 & 0 & -1 & 1 & 0 \\ 1 & 1 & 0 & -2 & -1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

$$R = \begin{pmatrix} 1 & 0 & 1 & 0 & 1 & -1 & 0 \\ 0 & 1 & -1 & -1 & 0 & 2 & 1 \end{pmatrix}.$$

The rows of R are the nonzero rows of the reduced echelon form. The 5 columns of N result from its free-variable equations; exact multiplication gives $AN = 0$, and their free coordinates show independence. Thus these are bases, not only spanning lists.

(b) Supply the transpose-kernel basis and check invertibility.

$$M = \begin{pmatrix} 0 & 2 & 0 & 0 \\ 1 & -1 & -1 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Here $A^T M = 0$, and its 4 columns are independent. Form $S = [\, R^T \; N \,]$ and $T = [\, C \; M \,]$ using the displayed blocks. There are $2 + 5 = 7$ columns in $S \in M_{7\times 7}$ and $2 + 4 = 6$ columns in $T \in M_{6\times 6}$. Exact determinants are

$$\det S = 23, \qquad \det T = -21.$$

Both are nonzero, verifying that both square matrices are invertible. Conceptually, a vector in both $\operatorname{Row} A$ and $\operatorname{Nul} A$ has form $x = A^T y$ and satisfies $Ax = 0$, so $x^T x = y^T Ax = 0$. Since $x^T x$ is a sum of squares, $x = 0$. The same argument with A^T treats $\operatorname{Col} A$ and $\operatorname{Nul} A^T$; their dimensions add to the full ambient dimensions. This explains why assembling their bases produces invertible matrices.

Exercise 4.6.37: Reconstruct the original matrix from pivot columns

Use the C and R displayed in Exercise 4.6.35. Those are precisely the pivot columns of the original A and the nonzero rows of its reduced echelon form. Multiplication gives

$$CR = \begin{pmatrix} 7 & -9 & -4 & 5 & 3 & -3 & -7 \\ -4 & 6 & 7 & -2 & -6 & -5 & 5 \\ 5 & -7 & -6 & 5 & -6 & 2 & 8 \\ -3 & 5 & 8 & -1 & -7 & -4 & 8 \\ 6 & -8 & -5 & 4 & 4 & 9 & 3 \end{pmatrix} = A.$$

For example, the third column of R is $(13/2, 11/2, 0, 0)$, so the third column of CR is $(13/2)a_1 + (11/2)a_2 = a_3$. The reduced matrix records the coefficients of every original column in terms of the pivot columns. Thus this product restores all columns exactly.

Exercise 4.6.38: Three generated rank factorizations and a general proof

For a reproducible family, use `random.Random(4600+r)` for each $r = 5, 4, 3$. Form C by stacking I_r over a $(5 - r) \times r$ array of successive `randint(-2,2)` draws, and form R by placing an $r \times (7 - r)$ array of subsequent draws after I_r. Set $A = CR$. The first r columns of A are C; its reduced echelon form has nonzero rows R. Thus these are actual examples of the exercise's construction, with exact rank r, rather than uncontrolled random ranks.

The rank-5 sample.

$$C = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \qquad R = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 2 & -1 \\ 0 & 1 & 0 & 0 & 0 & -1 & -2 \\ 0 & 0 & 1 & 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & -1 & 0 \end{pmatrix}.$$

$$CR = A = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 2 & -1 \\ 0 & 1 & 0 & 0 & 0 & -1 & -2 \\ 0 & 0 & 1 & 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & -1 & 0 \end{pmatrix}.$$

The rank-4 sample.

$$C = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -2 & 0 & 2 & 1 \end{pmatrix}, \qquad R = \begin{pmatrix} 1 & 0 & 0 & 0 & -1 & -1 & -1 \\ 0 & 1 & 0 & 0 & -2 & -1 & -1 \\ 0 & 0 & 1 & 0 & -2 & 0 & -1 \\ 0 & 0 & 0 & 1 & -1 & 0 & 2 \end{pmatrix}.$$

$$CR = A = \begin{pmatrix} 1 & 0 & 0 & 0 & -1 & -1 & -1 \\ 0 & 1 & 0 & 0 & -2 & -1 & -1 \\ 0 & 0 & 1 & 0 & -2 & 0 & -1 \\ 0 & 0 & 0 & 1 & -1 & 0 & 2 \\ -2 & 0 & 2 & 1 & -3 & 2 & 2 \end{pmatrix}.$$

The rank-3 sample.

$$C = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 2 & 2 & -2 \\ 1 & 2 & 2 \end{pmatrix}, \qquad R = \begin{pmatrix} 1 & 0 & 0 & 1 & 0 & -2 & 0 \\ 0 & 1 & 0 & 2 & -2 & 2 & -1 \\ 0 & 0 & 1 & 2 & -1 & -2 & 1 \end{pmatrix}.$$

$$CR = A = \begin{pmatrix} 1 & 0 & 0 & 1 & 0 & -2 & 0 \\ 0 & 1 & 0 & 2 & -2 & 2 & -1 \\ 0 & 0 & 1 & 2 & -1 & -2 & 1 \\ 2 & 2 & -2 & 2 & -2 & 4 & -4 \\ 1 & 2 & 2 & 9 & -6 & -2 & 0 \end{pmatrix}.$$

Prove the observed equality for every matrix.

Let C consist of the original pivot columns of a rank-r matrix A, and R of the nonzero rows of its reduced echelon form U. In U, the pivot columns are the first r coordinate vectors above the zero rows. Therefore the entries of column j of R express column j of U as a combination of its pivot columns. Elementary row operations preserve all column dependence relations: if $U = EA$ with E invertible, then $U\alpha = 0$ if and only if $A\alpha = 0$. The same coefficients thus express column j of A using columns of C. This means column j of CR is column j of A, for every j; hence $CR = A$.